



PROJECT DOCUMENTATION



Project Executable Files

 DATE
17 7/9/2026

 NAME
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 PROJECT
OptiCrop – Smart
Agricultural
Production
Optimization Engine

 MAX MARKS
3 Marks



Submission Checklist

ITEM TO SUBMIT	SUBMITTED
Complete source code (all files and folders)	Yes
README / Setup Guide	Yes
requirements.txt	Yes
Database schema / seed files (if applicable)	N/A
Environment configuration (.env.example)	No
Deployed application URL	No
APK / Executable binary	N/A
Dockerfile / Containerization config	No
Test files and test results	Yes
Demo video / walkthrough	Yes



File / Folder Structure

```
OPTICROP/  
├─ data/  
│   ├── Crop_recommendation.csv    # Dataset used for training  
│   └─ generate_dataset.py         # Script that builds the dataset  
├─ model/  
│   ├── train_model.py            # EDA, preprocessing, clustering, training  
│   ├── model.pkl                 # Trained Logistic Regression model  
│   └─ plots/                     # EDA charts + classification report  
├─ app/  
│   ├── app.py                   # Main Flask application  
│   ├── static/style.css          # CSS styling  
│   └─ templates/                 # base.html, home.html, about.html, findyourcrop.html  
├─ requirements.txt               # Python dependencies  
└─ README.md                     # Project documentation
```

FIELD	VALUE
Hosted / Deployed URL	Not Hosted (Runs Locally)
Login Credentials	Not required – no authentication in this version
Platform / Hosting Provider	Local Flask Server
Repository Link	https://github.com/hemanthkumar246/Opti_Crop/tree/main
Demo Video Link	[project demo video link]

 Run Instructions

- Clone the repository.
- Install dependencies: `pip install -r requirements.txt`
- (Optional) Regenerate the dataset and model: `python3 data/generate_dataset.py` then `python3 model/train_model.py`
- Run the app: `cd app && python3 app.py`
- Open `http://127.0.0.1:5000` in your browser.
- Go to “Find your crop”, enter soil and climate readings, and click Predict.

 Known Issues / Limitations

KNOWN ISSUE	WORKAROUND / STATUS
No recommendation history is stored	Can integrate a database
Runs locally only	Deploy to cloud
No user authentication	Can be added later